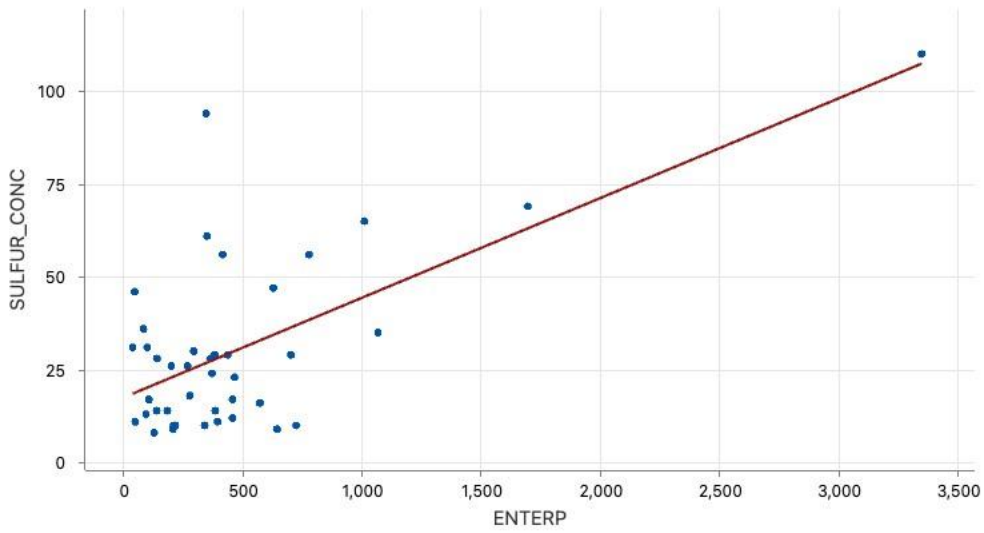
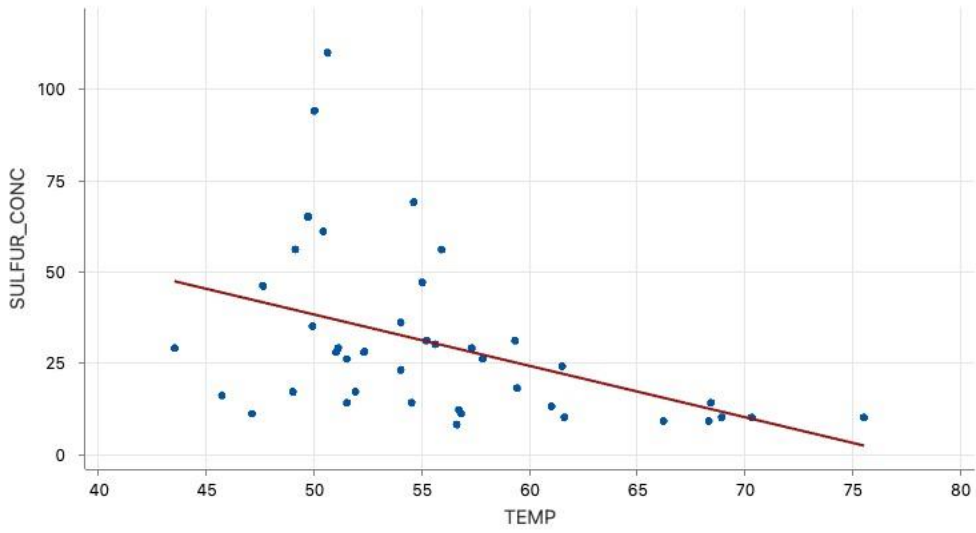
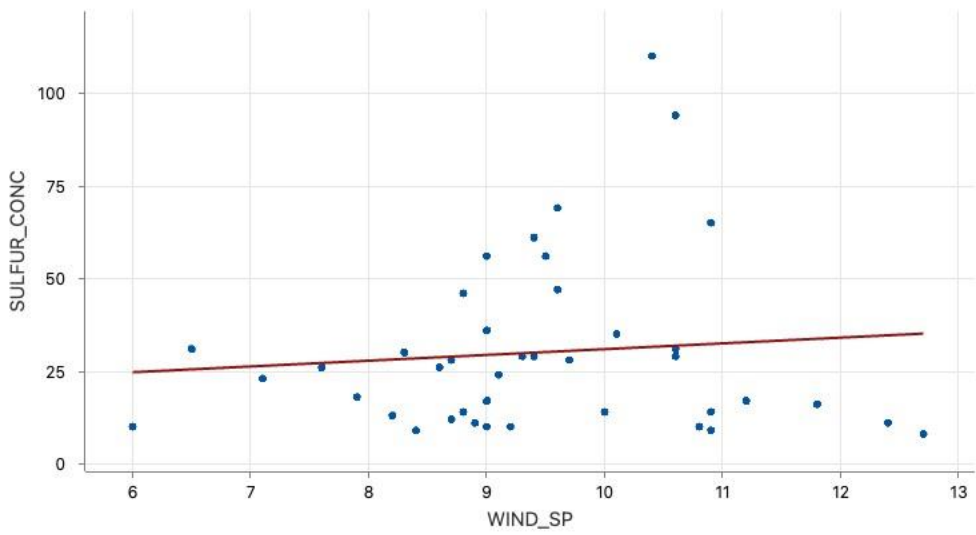
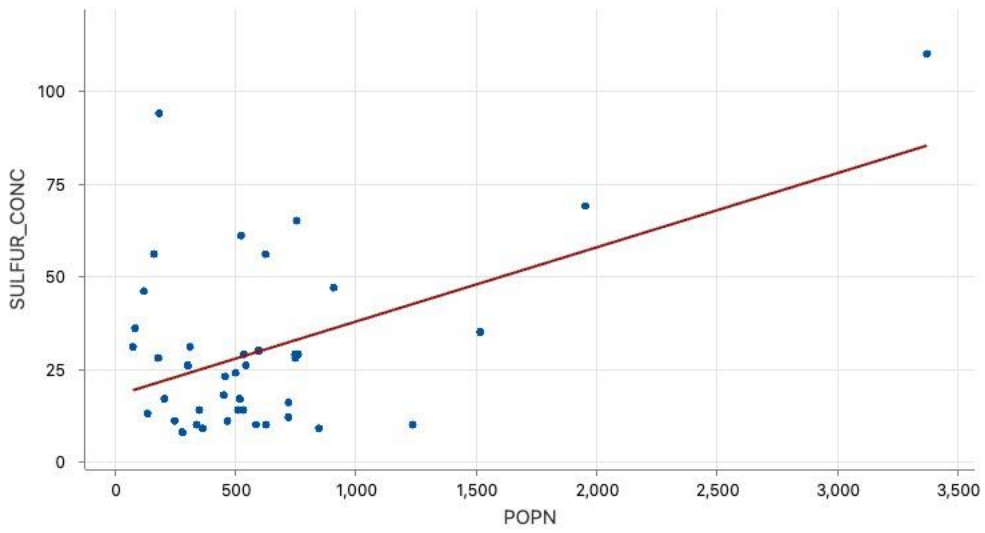
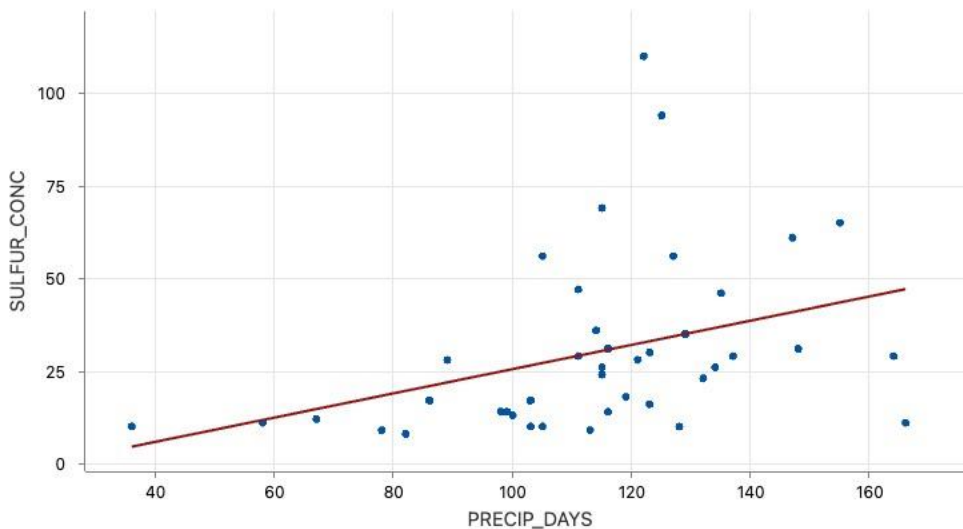
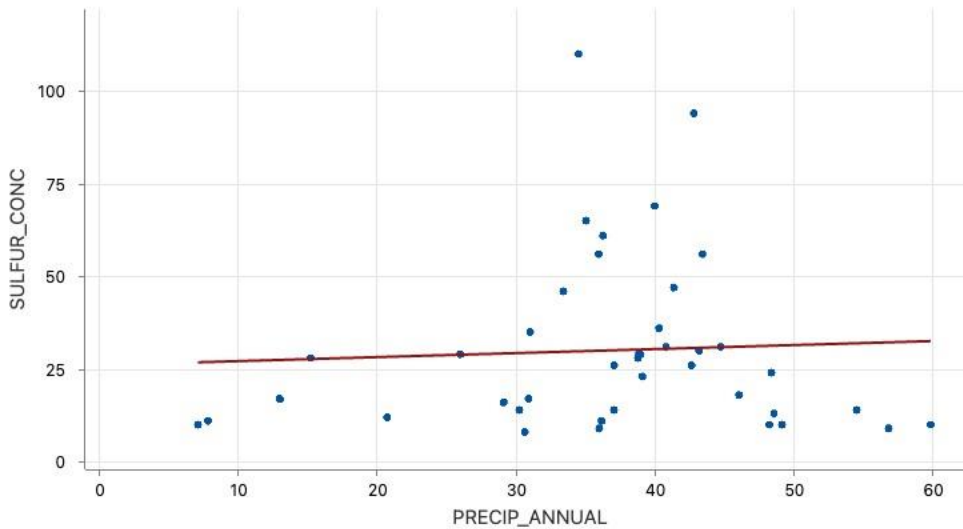


PROJECT FOR SULFUR DIOXIDE

1A)



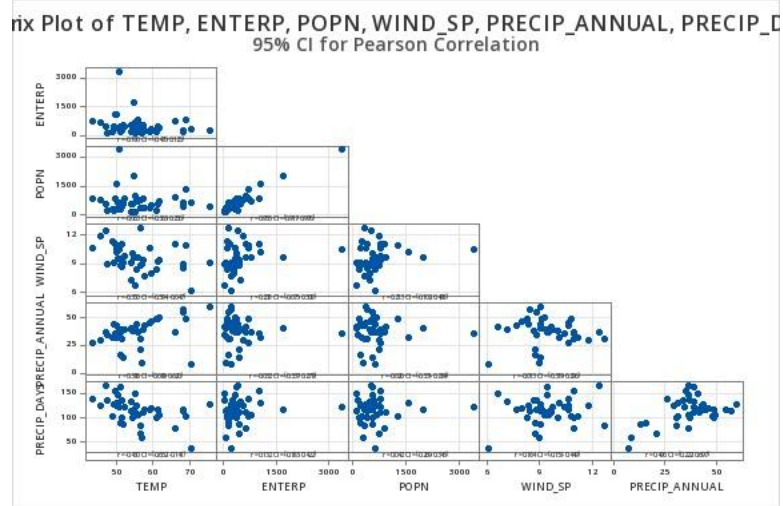
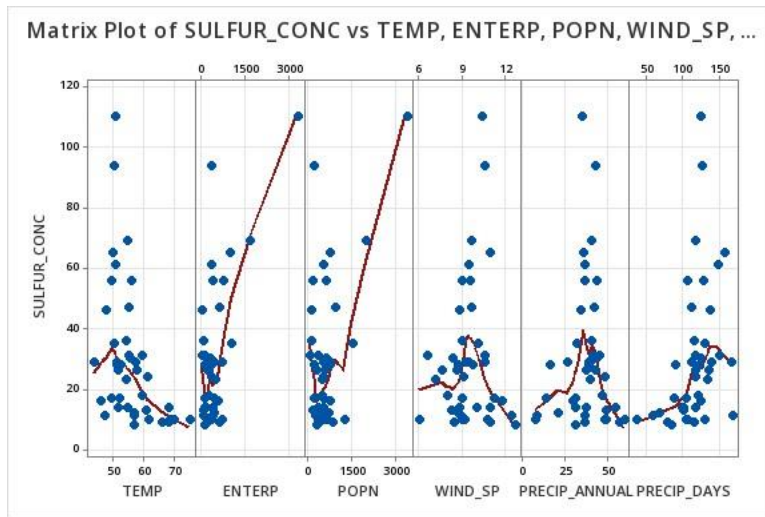




I will state my the following transformations, which the type of transformation is subjective. Between temperature (x) and Sulfur Concertation (y), a $1/x$ transformation is needed this is based on the graph pattern that I see got the curve going down. Between enterprises (x) and Sulfur Concertation (y), no transformation is needed and we can say that because based on the graph it is pretty linear. Between population size (x) and Sulfur Concertation (y), no transformation is needed and we can say that because based on the graph it is pretty linear. . Between wind speed (x) and Sulfur Concertation (y), a x^2 transformation is needed this is because based on the graph we can see a quadratic parabola pattern. Between precipitation annual (x) and Sulfur Concertation (y), a x^2 transformation is needed this is because based on the graph we can see a quadratic parabola pattern. Between precipitation days (x) and Sulfur Concertation (y), a $\ln(x)$ transformation is needed this is because based on the graph we can see a natural log pattern. Higher order terms will be needed for the precipitation annual (x) and wind speed (x).

1B)

WITHOUT TRANSFORMATIONS

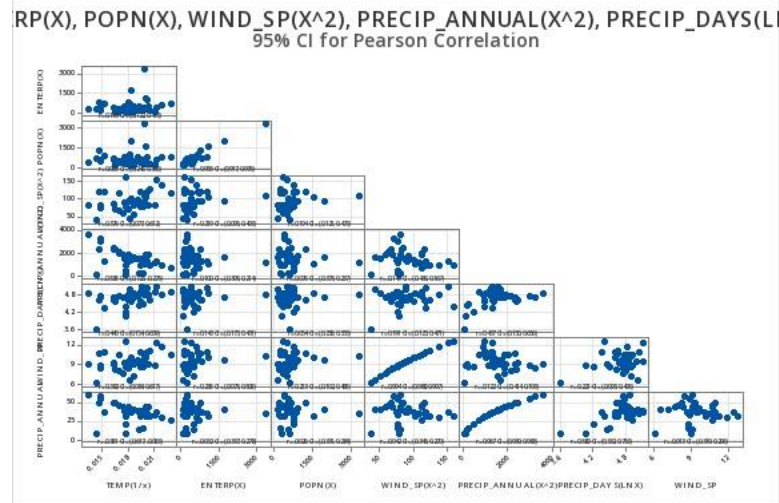


Correlations

TEMP ENTERP POPN WIND_SP PRECIP_ANNUAL

ENTERP	-0.190				
POPN	-0.063	0.955			
WIND_SP	-0.350	0.238	0.213		
PRECIP_ANNUAL	0.386	-0.032	-0.026	-0.013	
PRECIP_DAYS	-0.430	0.132	0.042	0.164	0.496

WITH TRANSFORMATIONS



Correlations

	TEMP(1/x)	ENTER P(X)	POP N(X)	WIND_SP(X^2)	PRECIP_ANNU AL(X^2)	PRECIP_DAY S(LNX)	WIND _SP
ENTERP(X)	0.193						
POP(X)	0.065	0.955					
WIND_SP(X^2)	0.376	0.219	0.194				
PRECIP_ANNU AL(X^2)	-0.538	-0.100	-0.076	-0.149			
PRECIP_DAYS(LN(X))	0.440	0.143	0.054	0.191	0.437		
WIND_SP	0.382	0.238	0.213	0.994	-0.122	0.221	
PRECIP_ANNU AL	-0.381	-0.032	-0.026	-0.042	0.967	0.580	-0.013

According to the Matrix Plot between our x-variables and y-variable, we can see that between Enterprise and Sulfur Concentration, Population Size and Sulfur Concentration you see that the curve deviates compared to the other variables. Which suggest that there is collinearity.

ENTERP(X) and POP(X) have high correlation coefficients which are (.955).

PRECIP_DAYS(LNX) and PRECIP_ANNUAL have a moderate correlation coefficient.

PRECIP_ANNUAL(X^2) and PRECIP_DAYS(LNX) have a moderate correlation coefficient.

PRECIP_ANNUAL(X^2) and PRECIP_ANNUAL have a high correlation coefficient.

WIND_SP(X^2) and WIND_SP have a high correlation coefficient. Most of the rest have relative low correlation coefficient.

1C)

WITHOUT TRANSFORMATIONS

Coefficients

Term	Coef	SE Coef	T-Value	P-Value	VIF
Constant	111.7	47.3	2.36	0.024	
TEMP	-1.268	0.621	-2.04	0.049	3.76
ENTERP	0.0649	0.0157	4.12	0.000	14.70
POP(X)	-0.0393	0.0151	-2.60	0.014	14.34
WIND_SP	-3.18	1.82	-1.75	0.089	1.26
PRECIP_ANNUAL	0.512	0.363	1.41	0.167	3.40
PRECIP_DAYS	-0.052	0.162	-0.32	0.750	3.44

Looking at the VIF for every variable, we see that the only variables with a VIF value greater than 10 are ENTERP, and POPN which means there is collinearity issues.

WITH TRANSFORMATIONS

Coefficients

Term	Coef	SE Coef	T-Value	P-Value	VIF
Constant	-86.2	88.2	-0.98	0.336	
TEMP(1/x)	3406	3115	1.09	0.282	8.56
ENTERP(X)	0.0678	0.0161	4.20	0.000	15.07
POPN(X)	-0.0433	0.0157	-2.76	0.010	15.13
WIND_SP(X^2)	-1.160	0.850	-1.36	0.182	97.17
PRECIP_ANNUAL(X^2)	-0.0114	0.0180	-0.64	0.529	37.18
PRECIP_DAYS(LNX)	-10.3	21.9	-0.47	0.641	7.00
WIND_SP	18.6	16.3	1.14	0.262	99.22
PRECIP_ANNUAL	1.21	1.15	1.05	0.302	33.53

Looking at the VIF for every variable, we see that the only variables with a VIF value greater than 10 are ENTERP, POPN, WIND_SP(X^2), PRECIP_ANNUAL(X^2), WIND_SP, PRECIP_ANNUAL, which means there is collinearity issues.

2A)

Response is SULFUR_CONC

P
R
EPCR
I E
PCPW_IR
I A P E
N N_ C
T E D ND I
E N _ U A P
MT P S A Y W_
P E OP L S I A
(R P (((N N
1PNXXLDN/((^ ^N_U
xXX22XSA
S)))))) P L

Vars	R-Sq	R-Sq (adj)	R-Sq (pred)	Mallovs	Cp			
1	41.6		40.1	37.8	21.7	18.170	X	
1	24.4	22.4	4.9	39.0	20.671		X	
2	58.6	56.5	53.0	6.6	15.489	X	X	
2	50.4	47.8	43.2	14.8	16.965	X X		
3	61.5	58.3	53.3	5.7	15.150	X	X	X
3	60.6	57.4	53.0	6.6	15.328	X X	X	

4	63.2	59.1	54.1	6.0	15.012	X	X	X	X										
4	63.2	59.1	53.7	6.0	15.019	X	X	X	X										X
5	66.3	61.5	55.7	4.9	14.570	X	X	X	X										X
5	65.8	60.9	55.2	5.3	14.669	X	X	X	X										
6	67.5	61.7	48.6	5.7	14.524	X	X	X	X									X	X
6	67.0		61.2		49.8			6.1	14.619	X	X	X	X	X				X	
7	67.9		61.1		45.4			7.2	14.636	X	X	X	X	X				X	X
7	67.7		60.9		45.2			7.4	14.678	X	X	X	X				X	X	X
8	68.1		60.2		40.8			9.0	14.812	X	X	X	X	X	X		X	X	X

2B)

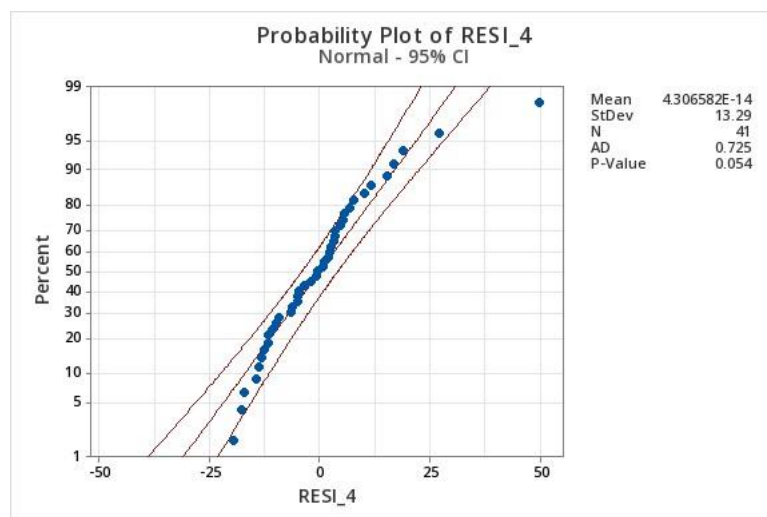
The best subset model that I pick is the first model with 7 variables. I picked this subset because the R-Sq is the highest (67.9) and adding 1 to the variables that it has ($k = 7 + 1$) is has a difference of .8 from the Mallows Cp which is pretty low.

2C)

Regression Equation

SULFUR_CONC = -109.5 + 2256 TEMP(1/x) + 0.0683 ENTERP(X) - 0.0441 POPN(X) - 1.182 WIND_SP(X^2) - 0.0123 PRECIP_ANNUAL(X^2) + 19.2 WIND_SP + 1.03 PRECIP_ANNUAL

3A)



3B)

Since the p-value from the probability plot is 0.054 which is more 0.05 which satisfied the normal distribution assumption.

3C)

Statistics

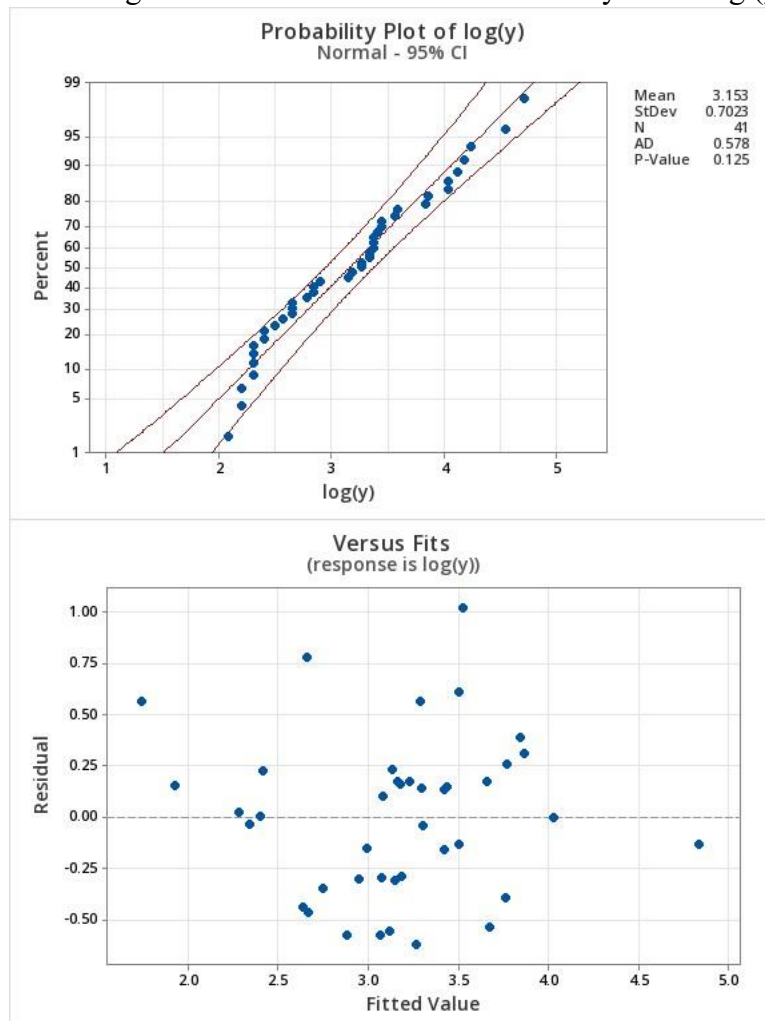
Variable	N	N*	Mean	SE Mean	StDev	Minimum	Q1	Median	Q3	Maximum
SULFUR_CONC	41	0	30.05	3.67	23.47	8.00	12.50	26.00	35.50	110.00

If we divided sqrt of mean (sqrt30.05) by SD (23.47) = .2336

If we divided the mean (30.05) by SD (23.47) = 1.2804

If we divided the squared mean (30.05^2) by SD (23.47) = 38.4748

According to our calculations above we can say that a log (y) transformation is needed on y.



Regression Equation

$$\begin{aligned} \log(y) = & -3.41 + 117.8 \text{ TEMP}(1/x) + 0.001432 \text{ ENTERP}(X) - 0.000951 \text{ POPN}(X) \\ & - 0.0567 \text{ WIND_SP}(X^2) - 0.000846 \text{ PRECIP_ANNUAL}(X^2) + 0.892 \text{ WIND_SP} \\ & + 0.0634 \text{ PRECIP_ANNUAL} \end{aligned}$$

Coefficients

Term	Coef	SE Coef	T-Value	P-Value	VIF
Constant	-3.41	2.11	-1.61	0.116	
TEMP(1/x)	117.8	55.8	2.11	0.042	3.29
ENTERP(X)	0.001432	0.000466	3.07	0.004	15.01
POPN(X)	-0.000951	0.000453	-2.10	0.043	14.99
WIND_SP(X^2)	-0.0567	0.0246	-2.31	0.027	96.86
PRECIP_ANNUAL(X^2)	-0.000846	0.000519	-1.63	0.112	36.84
WIND_SP	0.892	0.471	1.89	0.067	98.63
PRECIP_ANNUAL	0.0634	0.0316	2.01	0.053	30.07

Model Summary

S	R-sq	R-sq(adj)	R-sq(pred)
0.428605	69.27%	62.75%	47.09%

4A)

Fits and Diagnostics for Unusual Observations

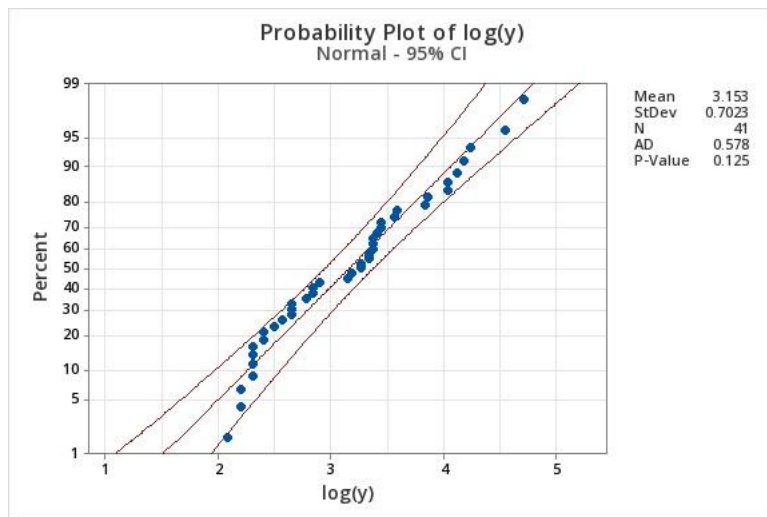
Obs	log(y)	Fit	SE Fit	95% CI	Resid	Std Resid	Del Resid	HI	Cook's D	DFITS		
1	2.303	1.738	0.340	(1.047, 2.430)	0.564	2.16	2.30	0.629099	0.99	2.99160	R	X
11	4.700	4.834	0.360	(4.101, 5.566)	-0.133	-0.57	-0.57	0.705793	0.10	-0.87876		X
31	4.543	3.523	0.161	(3.196, 3.849)	1.021	2.57	2.83	0.140415	0.13	1.14278	R	

R Large residual
X Unusual X

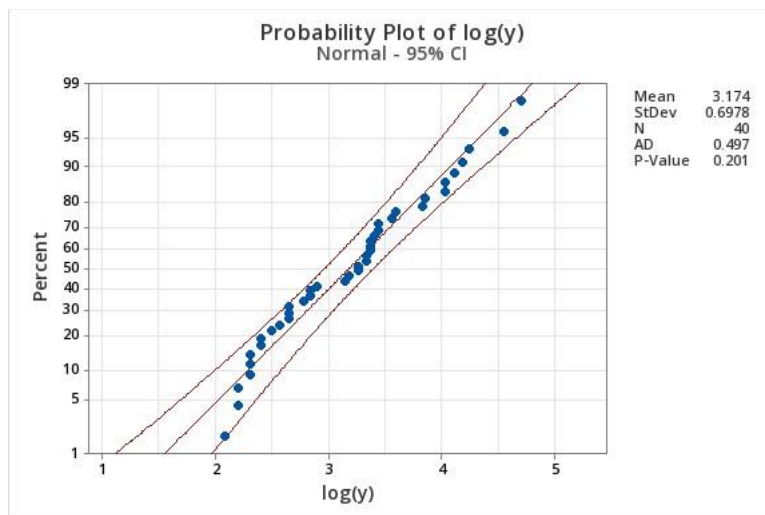
According to the table above the, the points that have high influence are observation 1 and observation 31, because the |DFITS| is greater than 1 makes it influential. Observation 31 have a weak leverage because HI is less than 0.2. Observation 1 and Observation 11 has high leverage because HI is more than 0.5

4B)

BEFORE



AFTER



As you can see the P-Value increased from 0.125 to 0.201 when we removed the 1st and 31st observation because it was influential. I also removed 11th observation because it had high leverage make it influential.

BEFORE

Model Summary

S	R-sq	R-sq(adj)	R-sq(pred)
0.428605	69.27%	62.75%	47.09%

AFTER

Model Summary

S	R-sq	R-sq(adj)	PRESS	R-sq(pred)	AICc	BIC
0.403271	72.59%	66.60%	8.22897	56.66%	55.94	65.14

As we see that R-sq increase from 69.27% to 72.59%, which might mean that it improve our model.

5A)

Regression Equation

$$\log(y) = -7.74 + 135.4 \text{ TEMP}(1/x) + 0.001517 \text{ ENTERP}(X) - 0.001096 \text{ POPN}(X) - 0.0927 \text{ WIND_SP}(X^2) - 0.001065 \text{ PRECIP_ANNUAL}(X^2) + 1.618 \text{ WIND_SP} + 0.0847 \text{ PRECIP_ANNUAL}$$

Plugging the given value we get

$$7.74 + 135.4(1/60) + 0.001517(150) - 0.001096(600) - 0.0927(100) - 0.001065(1600) + 1.618(10) + 0.0847(40) = 18.1606$$

Which implies that with the values provide below in the Settings tables the city of with value of temperature of 1/60, a value of enterprise of 150, a value of population size of 600(thousands), a wind speed of 10 mph, and squared wind speed of 10^2 mph, and a value of precipitation annual of 40(inches), and squared precipitation annual of 40^2 (inches) will result on a sulfur dioxide concentration of 18.1606.

Prediction

Fit	SE Fit	95% CI	95% PI
2.68205	0.166191	(2.34353, 3.02057)	(1.79359, 3.57051)

Settings

Variable	Setting
TEMP(1/x)	0.0167 ENTERP(X)
1600 WIND_SP	10 PRECIP_ANNUAL
	40
	150 POPN(X)
	600 WIND_SP(X^2)
	100 PRECIP_ANNUAL(X^2)

If we do a inverse of log transformation in y which is e^y . We get our 95% CI(10.1254, 20.1845) and our 95% PI is (4.8943, 31.8036).

5B)

CI Interpretation: We are 95% confident that for the true population of with value of temperature of 1/60, a value of enterprise of 150, a value of population size of 600(thousands), a wind speed of 10 mph, and squared wind speed of 10^2 mph, and a value of precipitation annual of 40(inches), and squared precipitation annual of 40^2 (inches) will result on a sulfur dioxide concentration between 10.1254 and 20.1845.

5C)

PI Interpretation: We are 95% confident that for a new single city with a value of temperature of 1/60, a value of enterprise of 150, a value of population size of 600(thousands), a wind speed of 10 mph, and squared wind speed of 10^2 mph, and a value of precipitation annual of 40(inches), and squared precipitation annual of 40^2 (inches) will result on a sulfur dioxide concentration between 4.8943 and 31.8036.